

Josh Richards - Product Engineer

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Skills

- Languages: TypeScript, C#, Go, Python, PostgreSQL
- Frameworks: React, Next.js, Astro, Tailwind, Node.js, .NET
- AI/Agentic: Claude Code, MCP, AGENTS.md, custom skills, software factory
- Platform/Ops: AWS, GCP, Docker, GitHub Actions, New Relic, Sumo Logic, WebSocket

Work Experience

[Xero - Accounting Software for Small Businesses - Product Engineer](#) - 2020/Present

Analytics (2025-Present) [TypeScript | Node.js | C# | GCP | PostgreSQL | New Relic | Observability | Agentic Tooling]

- Sole pre-existing Xero engineer in the new UK Analytics team formed to integrate Syft Analytics post-acquisition, bridging both orgs on identity, APIs, and incident process.
- Owned observability end-to-end, replacing lead-only GCP tooling with shared New Relic APM and dashboards so every engineer could trace and debug production independently.
- Built a production synthetic monitor from scratch (C# worker, AWS, OIDC, CI/CD, retries, Slack alerting), cutting incident detection from hours to minutes. It caught multiple incidents before any customer reports.
- Introduced MCP-driven agentic workflows and authored the team's AGENTS.md and agent skills, turning individual AI experimentation into team practice.
- Shipped the v3 Budgets API: resilience patterns, kill switch, and behavioural parity with the legacy accounting service.

Invoicing (2020-2025) [React | TypeScript | C# | ASP.NET | AWS | CI/CD]

- A core contributor on Xero's modernized Invoicing platform, driving the sunset of the legacy system and ensuring a smooth migration for over one million small businesses.
- Designed a recovery strategy that reduced mean time to recovery (MTTR) for the Invoicing frontend from 30 minutes to 30 seconds. Collaborated with my team's Lead Engineer to implement the approach and secured buy-in from the wider team.
- Won the People's Choice Award in Xero's annual hackathon for a beloved unicorn easter egg in Invoicing.
- Mentored and onboarded multiple new engineers into the team.

Personal Projects

[Fameboxd](#) - Astro | Claude Code | GitHub Actions | Tally | Cloudflare (2026) ([GitHub](#))

A directory of celebrities with public Letterboxd accounts. The content pipeline runs on agents: Tally form submission triggers custom Claude Code skill which validates, researches and writes each entry, a daily GitHub Action refreshes activity from RSS, and build-time validation rejects bad generated output before it ships. 3,000+ unique visitors with growing SEO.

[Album Listen List](#) - React | TypeScript | Tailwind | Supabase | Spotify OAuth | Vercel (2022) ([Reddit Post](#))

A Spotify companion for tracking albums to listen to, rating them, and sharing lists publicly. Live since 2022 and still growing organically: 11.4k unique visitors last year (+20%), 4 minute average sessions, now holding 30,000+ albums across 62,000 list entries. Migrated 1000+ users from Firebase to Supabase with zero downtime by moving each user's data on their next login. Redesigning the document store into a relational schema halved stored album rows and unlocked multi-list, public sharing, and per-album reviews.

[Ten Thousand Pixels](#) - JavaScript | Golang | WebSocket | Redis | Fly.io (2024) ([GitHub](#) / [Reddit Post](#))

A real-time collaborative pixel art canvas where users can draw pixels and see live updates. It gained over 20,000 unique visitors, 100 GitHub stars, and peaked at 30,000 pixel updates every 10 minutes.

Other

- Graduated with a Bachelor of Science in Computer Science from Victoria University of Wellington (2021)
- I stay active by bouldering a couple of times a week at my local gym, and I speed solve Rubik's Cubes competitively.